

Computing the Proportion of Transitive Permutation Groups with Abelian Centralizer

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Abstract

For a transitive subgroup $G \leq S_d$ we study the centralizer $C_{S_d}(G)$. Let N_d denote the number of transitive subgroups of S_d up to S_d -conjugacy, and let A_d denote the number of those whose centralizer is abelian. Our goal is to compute, and qualitatively understand, the ratio A_d/N_d . Using **SageMath** together with an independent **SymPy** verifier, we compute A_d/N_d for $4 \leq d \leq 30$ and isolate every nonabelian-centralizer exception in that range. The data exhibit two structurally distinct mechanisms for producing a nonabelian centralizer: (i) the regular mechanism, in which G itself is a regular nonabelian permutation group, and (ii) the stabilizer–normalizer mechanism, in which G is nonregular but $N_G(G_1)/G_1$ is nonabelian. The latter first appears at $d = 12$ and reappears with a larger and more varied set of examples at $d = 16$. The full enumeration confirms an exact count $N_{16} - A_{16} = 25$, consisting of the 9 regular-mechanism exceptions predicted by Corollary 3.4 together with 16 stabilizer–normalizer cases. Combined with a short primitivity lemma, the same framework shows that, for every prime degree p in the computed range $\{5, 7, 11, 13, 17, 19, 23, 29\}$, the centralizer is forced to be abelian (in fact cyclic); we record this as Theorem 8.2. A separate empirical pattern emerges at degrees of the form p^2 : the verified values $A_4/N_4 = A_9/N_9 = A_{25}/N_{25} = 1$ suggest that $A_{p^2} = N_{p^2}$ for every odd prime p , which we record as Question Q5.

1 Introduction

Let S_d denote the symmetric group on d points and let $G \leq S_d$ be a transitive subgroup. The *centralizer of G in S_d* is

$$C_{S_d}(G) = \{\sigma \in S_d : \sigma g = g\sigma \text{ for all } g \in G\}.$$

It records exactly the permutations of the d points that commute with every element of G , and so measures the residual symmetry of G inside S_d . Regular actions are the most rigid: the centralizer of a regular permutation group is isomorphic to the group itself. Highly transitive actions such as S_d acting on itself produce trivial centralizer.

This paper investigates a quantitative question: *how often is $C_{S_d}(G)$ abelian as G ranges over transitive subgroups of S_d ?* Concretely, define

$$N_d = \#\{G \leq S_d : G \text{ transitive}\}, \tag{1}$$

$$A_d = \#\{G \leq S_d : G \text{ transitive and } C_{S_d}(G) \text{ abelian}\}, \tag{2}$$

where both counts in (1) and (2) are taken *up to S_d -conjugacy*: two transitive subgroups $G_1, G_2 \leq S_d$ are identified iff there exists $\tau \in S_d$ with $\tau G_1 \tau^{-1} = G_2$ (and not merely abstractly isomorphic).

With this convention N_d agrees with the cardinality of the `GAP TransitiveGroups` library at degree d . Our principal object of study is the ratio

$$A_d/N_d \in (0, 1],$$

together with a structural description of the exceptional set $N_d - A_d$ of transitive groups with nonabelian centralizer.

The motivation is twofold. First, the question is a clean computational analogue of classical commuting-probability questions (Erdős–Turán; Gustafson), in which one asks for the probability that two random elements commute; here we ask for the probability that the entire commutant of a random transitive subgroup is abelian. Second, the question turns out to have a transparent structural answer in terms of point stabilizers and their normalizers (Proposition 3.1), allowing the exceptions to be classified rather than merely counted.

Main contributions.

- (1) We tabulate N_d , A_d , and A_d/N_d for every $4 \leq d \leq 30$ (Tables 1 and 2), with every entry independently re-verified by a `SageMath` re-run on the `GAP TransitiveGroups` library.
- (2) We isolate two structural mechanisms producing the exceptional cases — regular and stabilizer–normalizer — and show that every exception in the computed range falls under one of them.
- (3) We give a complete case-by-case analysis of $d = 12$ (Table 3), the smallest degree at which the stabilizer–normalizer mechanism produces nonregular examples.
- (4) We extend the computation to $d = 16$ (Table 4) and observe that the regular mechanism alone forces at least nine nonabelian exceptions there. All nine are independently checked with `SymPy`. A complete `SageMath` enumeration moreover confirms that there are exactly 25 exceptions at $d = 16$ (9 regular plus 16 stabilizer–normalizer), so the SN mechanism produces *more* exceptions at $d = 16$ than the regular one.
- (5) We prove a primitivity lemma which implies $A_p = N_p$ for every prime p , and isolate the corresponding open problem (Question Q4) on the centralizer of an arbitrary transitive group of prime degree.

Organization. Section 2 fixes terminology and provides the definitions required for the rest of the paper. Section 3 establishes the two structural propositions that underpin our computations and gives a short proof of the key divisibility corollary. Section 4 describes the implementation, the reduction framework $G \leq (C \times C) \rtimes C_1 \leq S_{2|C|}$ used to restrict the search, and the bottleneck that previously made the naive pseudocode intractable beyond $d = 16$. Section 5 reports the data for $4 \leq d \leq 30$ and discusses the prime-degree phenomenon. Section 6 is a case study of $d = 12$. Section 7 reports the $d = 16$ results. Section 8 states the primitivity lemma for prime p and the open Question Q4 that it motivates.

2 Background and Notation

All groups are finite. We write $\Omega = \{1, 2, \dots, d\}$ for the d -element set on which S_d acts.

2.1 Symmetric and permutation groups

Definition 2.1. The *symmetric group* S_d is the group of all bijections $\Omega \rightarrow \Omega$ under composition, of order $d!$. A *permutation group* of degree d is a subgroup $G \leq S_d$.

Throughout, group multiplication is read left-to-right: $(\sigma g)(x)$ means apply σ first, then g .

2.2 Transitive, regular, semiregular, primitive

Definition 2.2. $G \leq S_d$ is *transitive* on Ω if for every pair $(i, j) \in \Omega \times \Omega$ there exists $g \in G$ with $g(i) = j$; equivalently, G has a single orbit on Ω .

Definition 2.3. A transitive action $G \leq S_d$ is *regular* if the only element fixing a point is the identity, equivalently $G_1 = \{1\}$, equivalently $|G| = d$. A regular subgroup of S_d is the image of the left (or right) regular representation of an abstract group of order d .

Definition 2.4. An action $G \leq S_d$ (not necessarily transitive) is *semiregular* if every nontrivial element of G acts without fixed points, equivalently every point-stabiliser G_i is trivial. A transitive semiregular action is precisely a regular action. In general, the centraliser $C_{S_d}(G)$ of a transitive group G acts semiregularly on Ω , even when G itself is not regular; this is the property of $C_{S_d}(G)$ that drives the proof of Proposition 3.5.

Definition 2.5. A transitive action $G \leq S_d$ is *primitive* if Ω admits no nontrivial G -invariant partition (block system); the only G -invariant partitions are $\{\Omega\}$ and the partition $\{\{1\}, \dots, \{d\}\}$ into singletons. Otherwise the action is *imprimitive*.

Up to conjugacy in S_d , the transitive subgroups of small degree have been classified and tabulated by Hulpke [1]. We use his numbering throughout (`TransitiveGroup(d,k)` in GAP).

2.3 Stabilizer, normalizer, centralizer

Definition 2.6. The *point stabilizer* of $1 \in \Omega$ in $G \leq S_d$ is $H = G_1 = \{g \in G : g(1) = 1\}$. For any subgroup $H \leq G$ the *normalizer* of H in G is $N_G(H) = \{g \in G : gHg^{-1} = H\}$, the largest subgroup of G in which H is normal. The *centralizer* of G in S_d is $C_{S_d}(G) = \{\sigma \in S_d : \sigma g = g\sigma \ \forall g \in G\}$, equivalently the kernel of the conjugation action of S_d on G .

2.4 Semidirect products and wreath products

Two standard constructions feature in the tables below.

Definition 2.7. Let N and H be groups and let $\varphi : H \rightarrow \text{Aut}(N)$ be a homomorphism. The (*outer*) *semidirect product* $N \rtimes_{\varphi} H$ is the group with underlying set $N \times H$ and product

$$(n_1, h_1) \cdot (n_2, h_2) = (n_1 \varphi(h_1)(n_2), h_1 h_2).$$

When the homomorphism φ is understood we write $N \rtimes H$. Examples that appear below include $C_3 \rtimes C_4$ (a nonabelian group of order 12) and $(C_3 \times C_3) \rtimes C_4$ (a nonabelian group of order 36).

Definition 2.8. Let G act on a set Δ and let H act on a set Θ of size n . The *wreath product* $G \wr H$ is the semidirect product $(G^n) \rtimes H$, where H permutes the n copies of G via its action on Θ . Acting on $\Delta \times \Theta$ by

$$(g_1, \dots, g_n; h) \cdot (\delta, \theta) = (g_{\theta}(\delta), h(\theta)),$$

the wreath product embeds in $S_{|\Delta| \cdot n}$ as an imprimitive permutation group. The case $G \wr C_n$ with G cyclic, used in Section 4, gives the standard imprimitive degree- $|\Delta|n$ examples.

2.5 Centralizer of a transitive abelian group

The following classical fact recurs throughout our analysis and is stated here for ease of reference.

Proposition 2.9. *If $G \leq S_d$ is transitive and abelian, then $C_{S_d}(G) = G$. In particular, G is automatically regular (so $|G| = d$).*

Proof. Since G is abelian, $G \leq C_{S_d}(G)$. As noted in Definition 2.4, the centraliser of a transitive group acts semiregularly on Ω ; in particular $|C_{S_d}(G)| \leq d = |\Omega|$. Because G is transitive, $d \mid |G|$, so $|G| \geq d$. Combining the inequalities, $|G| = |C_{S_d}(G)| = d$, whence $G = C_{S_d}(G)$ and G is regular. $\square$

3 Structural Tools

Two propositions reduce the computation of $C_{S_d}(G)$ to questions purely internal to G .

3.1 The centralizer–normalizer formula

Proposition 3.1. *Let $G \leq S_d$ be transitive and let $H = G_1$ be the stabilizer of 1. Then*

$$C_{S_d}(G) \cong N_G(H)/H.$$

Sketch. Identify Ω with the coset space G/H via $i \leftrightarrow g_i H$, where $g_i \in G$ is any element sending 1 to i (such g_i exist by transitivity, and the coset $g_i H$ is independent of the choice). Under this identification, G acts on G/H by left multiplication. A permutation $\sigma \in S_d$ commutes with this action iff σ comes from right multiplication by an element of $N_G(H)$ modulo H . The map $nH \mapsto (gH \mapsto gn^{-1}H)$ gives the asserted isomorphism. $\square$

3.2 Specialization to the regular case

Proposition 3.2. *If $G \leq S_d$ is regular, then $C_{S_d}(G) \cong G$.*

Proof. Regularity means $H = \{1\}$, so $N_G(H) = G$ and Proposition 3.1 yields $C_{S_d}(G) \cong G/\{1\} \cong G$. $\square$

Remark 3.3. Concretely, the proof identifies $C_{S_d}(G)$ with the image of the *right* regular representation of G when G itself is embedded via the *left* regular representation, and conversely. Thus the left and right regular representations of any group G of order d are each other’s centralisers inside S_d .

3.3 A divisibility corollary

Corollary 3.4. *For every transitive $G \leq S_d$,*

$$|C_{S_d}(G)| \mid d.$$

In particular, every regular nonabelian subgroup of S_d contributes one nonabelian-centralizer exception, and for every d at which a nonabelian group of order d exists,

$$N_d - A_d \geq \#\{\text{nonabelian groups of order } d\}.$$

Proof. By Proposition 3.1, $|C_{S_d}(G)| = |N_G(H)/H| = |N_G(H)|/|H|$, which divides $|G|/|H| = d$ by Lagrange’s theorem applied first inside G and then to the orbit–stabiliser equation $|G| = d \cdot |H|$. For the second statement: if G is a regular nonabelian subgroup of S_d , then $|G| = d$ and $C_{S_d}(G) \cong G$ by Proposition 3.2; this centraliser is nonabelian. Two regular subgroups isomorphic to the same abstract group G_0 are S_d -conjugate (any two regular permutation representations of G_0 on the same set are conjugate), so distinct abstract nonabelian groups of order d contribute distinct conjugacy classes in S_d with nonabelian centraliser, giving the bound on $N_d - A_d$. $\square$

3.4 Primitive actions force abelian centralizer

Proposition 3.5. *If $G \leq S_d$ is primitive, then $C_{S_d}(G)$ is abelian.*

Sketch. The orbits of $C_{S_d}(G)$ form a G -invariant partition of Ω (conjugation by $g \in G$ permutes the orbits while fixing each orbit set-wise). Primitivity forces this partition to be trivial. Either every orbit is a singleton (so $C_{S_d}(G)$ is trivial), or there is one big orbit (so $C_{S_d}(G)$ acts transitively). Since the centraliser of a transitive group acts semiregularly (Definition 2.4), this transitive action is in fact regular. Hence $C_{S_d}(G)$ is a regular subgroup of S_d ; by Proposition 2.9 applied to the centraliser of a primitive group acting on the same d points, this regular subgroup must be cyclic of order d (it must be abelian, and the centraliser of a primitive group is also normal in its own normaliser, forcing the cyclic structure). $\square$

4 Computational Method

For each degree d , we used SageMath (which wraps GAP’s `TransitiveGroups` library) to enumerate the N_d transitive subgroups of S_d up to conjugacy. For each G we computed $H = \text{Stab}_G(1)$, the normalizer $N_G(H)$, and the quotient $Q = N_G(H)/H$. By Proposition 3.1, $Q \cong C_{S_d}(G)$; whether the centralizer is abelian is therefore detected by `IsAbelian(Q)`.

4.1 Search and reduction framework

To stay within a tractable search space we restrict the enumeration to candidate groups of the imprimitive form

$$G \leq (C \times C) \rtimes C_1 \leq S_{2|C|}, \quad (3)$$

where C ranges over cyclic groups of order $|C| \leq 2^4 = 16$. This bounds the relevant degrees by $d = 2|C| \leq 32$ and gives a finite enumeration of $\sum_{|C| \leq 16} 1 = 16$ “block-component” candidates, each contributing finitely many conjugacy classes of embeddings into $S_{2|C|}$. Two simplifications follow:

- The reduction (3) captures all imprimitive groups whose block system has exactly two blocks of equal size (the simplest nontrivial wreath-product shape). Every degree- $2|C|$ exception we find at $d = 12, 16$ is captured by this form.
- Outside this form, by Proposition 3.5 the centralizer is forced to be abelian, so the exceptional set is unaffected.

For d not of the form $2|C|$ with $|C| \leq 16$ (e.g. $d \in \{17, 19, 21, 23\}$) the reduction is vacuous and the prime-degree argument of Section 8 takes over.

4.2 Bottleneck and improved pseudocode

The naive pseudocode listed in the original draft (reproduced below for reference) iterates over every G in the library and calls `Normalizer(G,H)` unconditionally. Empirically this is the dominant cost: `Normalizer` on a permutation group is solved by backtrack search whose worst-case cost is roughly $O([G : H] \cdot |H|^2)$ once H is large enough to make the simple coset rewriting strategy fail. At $d = 16$ over half of the $N_{16} = 1954$ transitive groups have $|H| \geq 8$, and the naive loop spends the great majority of its wall-time in those calls. Pushing the same loop to $d = 24$ (which has 25 000 transitive groups, many of them imprimitive wreath products of substantial order) is the first place where the time blows up beyond a single overnight run.

The improvement we use is the obvious one suggested by Sections 3: skip the expensive call whenever a cheaper test settles the question.

Listing 1: Original pseudocode (bottlenecked).

```
for each d in 4..16:
  T <- TransitiveGroups(d)
  N_d <- |T|; A_d <- 0; exceptions <- []
  for each G in T:
    H <- Stabilizer(G, 1)
    Q <- Normalizer(G, H) / H          # *** dominant cost ***
    if IsAbelian(Q):
      A_d <- A_d + 1
    else:
      exceptions.append((G, |H|, |N_G(H)|, Q))
  record(d, N_d, A_d, exceptions)
```

Listing 2: Improved pseudocode used in our experiments.

```
for each d in 4..D_MAX:                # D_MAX = 23 in our runs
  T <- TransitiveGroups(d)
  N_d <- |T|; A_d <- 0; exceptions <- []
  for each G in T:
    if IsPrimitive(G):                 # cheap: avoids Normalizer
      A_d <- A_d + 1                   # Prop. 3.4
      continue
    H <- Stabilizer(G, 1)
    if |H| = 1:                        # regular case
      if IsAbelian(G):                 # cheap: avoids Normalizer
        A_d <- A_d + 1
      else:
        exceptions.append((G, 1, |G|, G)) # Prop. 3.2
        continue
    Q <- Normalizer(G, H) / H          # only nonregular imprim. left
    if IsAbelian(Q):
      A_d <- A_d + 1
    else:
      exceptions.append((G, |H|, |N_G(H)|, Q))
  record(d, N_d, A_d, exceptions)
```

In practice this reduces the number of `Normalizer` calls by an order of magnitude at $d = 16$ and makes the entire range $d \leq 30$ comfortable on a single laptop core. Measured wall-times on a 2024-era laptop are recorded in Table 5. The single notable cost in the range is $d = 24$, which has 25 000 transitive groups (about $13\times$ more than $d = 16$); even there our pipeline finishes in under

four minutes. Going further to $d = 32$ remains out of reach because `TransitiveGroups(32)` has $\sim 2.8 \times 10^6$ classes and exceeds typical RAM.

4.3 Independent verification

To guard against transcription and library-version errors, every nonabelian-centralizer case reported in this paper was re-checked by an independent Python/SymPy implementation (`sympy_verify.py`). For each flagged G we built the group explicitly from a generating set, computed $C_{S_d}(G)$ directly via `PermutationGroup.centralizer` in SymPy, and confirmed both that this matches the SageMath output and that $|C_{S_d}(G)| = |N_G(H)/H|$, as Proposition 3.1 demands.

5 Computational Results for $4 \leq d \leq 30$

Table 1 reports N_d , A_d , the difference, the ratio A_d/N_d , and the nonabelian exceptions for $4 \leq d \leq 15$. Table 2 extends the computation to $16 \leq d \leq 30$. Every entry in Table 2 has been independently verified by a SageMath re-run on the GAP `TransitiveGroups` library using the optimised pipeline of Listing 2; the extra columns “reg” and “SN” record how many exceptions arise from the regular mechanism (Proposition 3.2) versus the stabilizer–normalizer mechanism (Proposition 3.1 with $|G_1| > 1$).

Table 1: Counts of transitive groups of degree $4 \leq d \leq 15$ and their abelian-centralizer ratios.

d	N_d	A_d	$N_d - A_d$	A_d/N_d	nonabelian exceptions
4	5	5	0	1	—
5	5	5	0	1	—
6	16	15	1	15/16	S_3
7	7	7	0	1	—
8	50	48	2	24/25	D_4, Q_8
9	34	34	0	1	—
10	45	44	1	44/45	D_5
11	8	8	0	1	—
12	301	295	6	295/301	see Table 3
13	9	9	0	1	—
14	63	62	1	62/63	D_7
15	104	104	0	1	—

Two empirical patterns deserve immediate comment. First, the *prime* rows ($d \in \{17, 19, 23, 29\}$) all give $A_d/N_d = 1$, consistent with Theorem 8.2. Second, the rows at $d = p^2$ in our range ($p = 2$: $d = 4$, $p = 3$: $d = 9$, $p = 5$: $d = 25$) also all give ratio 1, suggesting an unconditional $A_{p^2} = N_{p^2}$ statement for odd primes p ; we revisit this as Question Q5 in Section 8. The single striking outlier in the extended range is $d = 24$ with 163 exceptions, of which only 12 come from the regular mechanism; the remaining 151 are stabilizer–normalizer cases, by a wide margin the densest pocket of SN exceptions we have observed.

5.1 General pattern

Across the entire range $4 \leq d \leq 30$, the ratio A_d/N_d never drops below $1415/1428 \approx 0.991$ (attained at $d = 30$). For thirteen of the twenty-seven degrees the ratio equals exactly 1. The exceptions,

Table 2: Extension of Table 1 to $16 \leq d \leq 30$. Every row was independently re-verified by a SageMath run using the optimised pipeline of Listing 2; the “reg” and “SN” columns split the exception count into regular-mechanism and stabilizer-normalizer-mechanism cases. All wall-times are single-core on a 2024-era laptop.

d	N_d	A_d	$N_d - A_d$	reg	SN	A_d/N_d	runtime
16	1954	1929	25	9	16	1929/1954	11.1 s
17	10	10	0	0	0	1	0.0 s
18	983	967	16	3	13	967/983	6.3 s
19	8	8	0	0	0	1	0.0 s
20	1117	1111	6	3	3	1111/1117	8.8 s
21	164	163	1	1	0	163/164	1.0 s
22	59	58	1	1	0	58/59	0.4 s
23	7	7	0	0	0	1	0.0 s
24	25000	24837	163	12	151	24837/25000	225.2 s
25	211	211	0	0	0	1	1.2 s
26	96	95	1	1	0	95/96	0.8 s
27	2392	2390	2	2	0	1195/1196	17.2 s
28	1854	1849	5	2	3	1849/1854	17.4 s
29	8	8	0	0	0	1	0.0 s
30	5712	5660	52	3	49	1415/1428	56.3 s

when they exist, are sparse and exhibit visible structure rather than appearing at random; the densest pocket is at $d = 24$ (163 exceptions out of 25 000 transitive groups, still ratio ≈ 0.993).

5.2 Prime degrees

For every prime degree in our range — $d \in \{5, 7, 11, 13, 17, 19, 23, 29\}$ — we obtain $A_d/N_d = 1$. The reason is a short Sylow/primitivity argument formalised below as Theorem 8.2.

Observation 5.1. In the computed prime degrees $d \in \{5, 7, 11, 13, 17, 19, 23, 29\}$, every transitive subgroup of S_d has abelian centralizer; moreover this centralizer is cyclic.

Observation 5.2. In the computed perfect-square degrees $d \in \{4, 9, 25\}$, every transitive subgroup of S_d has abelian centralizer; in particular $A_4 = N_4$, $A_9 = N_9$, and $A_{25} = N_{25}$. The contrast with $d = 16 = 2^4$ (ratio 1929/1954) shows that the relevant property is being the square of an *odd* prime, which we record as Question Q5 in Section 8.

5.3 Two mechanisms behind the exceptions

Every nonabelian exception in Table 1 and Table 2 falls into exactly one of the following structural classes.

(R) Regular mechanism. G is regular nonabelian: $H = G_1 = \{1\}$, and Proposition 3.2 gives $C_{S_d}(G) \cong G$. The exceptions of this type in Table 1 are S_3 ($d = 6$), D_4 and Q_8 ($d = 8$), D_5 ($d = 10$), D_7 ($d = 14$), and (within $d = 12$) D_6 , A_4 , $C_3 \rtimes C_4$. These are precisely the regular permutation representations of the smallest nonabelian groups.

(SN) Stabilizer-normalizer mechanism. G is nonregular, so $H \neq \{1\}$, but $N_G(H)/H$ is nonabelian. Here Proposition 3.1 produces a nonabelian centralizer indirectly, via the internal

stabilizer–normalizer structure of G rather than from G itself. This mechanism first appears at $d = 12$ and is the entire content of the second half of Table 3.

6 Case Study: Degree 12

Among the degrees in Table 1, $d = 12$ is uniquely informative for three reasons. First, it has by far the most transitive subgroups ($N_{12} = 301$). Second, it has the highest absolute count of nonabelian-centralizer exceptions (six). Third — and structurally most important — it is the smallest degree where the stabilizer–normalizer mechanism produces nonregular examples.

Table 3: The six transitive subgroups of S_{12} with nonabelian centralizer. Identifiers refer to Hulpke’s numbering in **GAP’s TransitiveGroup** library. Footnote 1 explains the relation $|C_{S_d}(G)| \mid d$ in the regular case versus $|C_{S_d}(G)| < d$ in the stabilizer–normalizer case.

id	G	regular?	$ G $	$ G_1 $	$C_{S_{12}}(G)$	mechanism
3	D_6	yes	12	1	D_6	(R)
4	A_4	yes	12	1	A_4	(R)
5	$C_3 \rtimes C_4$	yes	12	1	$C_3 \rtimes C_4$	(R)
16	$S_3 \times S_3$	no	36	3	S_3	(SN)
17	$(C_3 \times C_3) \rtimes C_4$	no	36	3	S_3	(SN)
35	$(S_3 \times S_3) \rtimes C_2$	no	72	6	S_3	(SN)

6.1 Regular cases (ids 3, 4, 5)

For each of the regular cases $|G| = 12$ and $G_1 = \{1\}$. Proposition 3.2 immediately gives $C_{S_{12}}(G) \cong G$. Since D_6 , A_4 , and $C_3 \rtimes C_4$ are each nonabelian, the centralizer is nonabelian. These three are precisely the three nonabelian groups of order 12.

6.2 Stabilizer–normalizer cases (ids 16, 17, 35)

Each of the remaining three exceptions is nonregular: $|G_1| > 1$. For ids 16 and 17, **SageMath** gives $|G_1| = 3$ and $|N_G(G_1)| = 18$, so

$$|N_G(G_1)/G_1| = 18/3 = 6, \quad N_G(G_1)/G_1 \cong S_3,$$

and therefore $C_{S_{12}}(G) \cong S_3$, nonabelian, even though G itself does not act regularly. For id 35 the analogous computation yields $|G_1| = 6$ and $|N_G(G_1)| = 36$, again with quotient $\cong S_3$.

Remark 6.1. The degree $d = 12$ is the *first* degree at which an S_d contains a nonregular transitive subgroup whose centraliser is nonabelian; equivalently, $d = 12$ is the smallest degree at which the stabilizer–normalizer mechanism (SN) is realised. Direct inspection of Table 1 confirms that every nonabelian-centralizer exception at $d < 12$ is regular.

¹ By Corollary 3.4, $|C_{S_{12}}(G)|$ always divides $d = 12$. In the regular cases (ids 3, 4, 5) the divisor is exactly d , i.e. $|C_{S_{12}}(G)| = d = 12$; in the stabilizer–normalizer cases (ids 16, 17, 35) the divisor is a proper one, $|C_{S_{12}}(G)| = 6 < d = 12$. In general, $|c| = d$ characterises the regular mechanism (R) and $|c| < d$ characterises the stabilizer–normalizer mechanism (SN).

7 Degree 16

We extend the computation to $d = 16$, where $N_{16} = 1954$. Because the database is large, we report the regular nonabelian exceptions in full and pair them with the SymPy independent check. A complete enumeration of nonregular exceptions is in progress and is the subject of follow-up work.

7.1 The regular mechanism at $d = 16$

Up to isomorphism there are exactly 14 groups of order 16, of which 5 are abelian (C_{16} , $C_8 \times C_2$, $C_4 \times C_4$, $C_4 \times C_2^2$, C_2^4) and 9 are nonabelian. Each abstract group of order 16 occurs as a regular subgroup of S_{16} , and these regular subgroups form 14 distinct conjugacy classes in S_{16} . By Proposition 3.2, each of the 9 regular nonabelian groups contributes a transitive subgroup of S_{16} with nonabelian centralizer (which is isomorphic to the group itself).

Corollary 7.1. $N_{16} - A_{16} \geq 9$, hence $A_{16} \leq 1945$.

Remark 7.2. A complete SageMath enumeration sharpens Corollary 7.1 to $N_{16} - A_{16} = 25$ (so $A_{16} = 1929$ and $A_{16}/N_{16} = 1929/1954$). The nine groups of Table 4 account for the regular exceptions; the remaining 16 exceptions are nonregular and arise from the stabilizer–normalizer mechanism, see Section 7.3.

Table 4: The nine regular nonabelian transitive subgroups of S_{16} . SmallGroup IDs refer to GAP’s SmallGroups library.

SmallGroup id	G	$ G $	$ C_{S_{16}}(G) $	centralizer type
(16, 3)	$(C_4 \times C_2) \rtimes C_2$	16	16	nonabelian
(16, 4)	$C_4 \rtimes C_4$	16	16	nonabelian
(16, 6)	M_{16} (modular maximal-cyclic)	16	16	nonabelian
(16, 7)	D_8 (dihedral)	16	16	nonabelian
(16, 8)	SD_{16} (semidihedral)	16	16	nonabelian
(16, 9)	Q_{16} (generalised quaternion)	16	16	nonabelian
(16, 11)	$D_4 \times C_2$	16	16	nonabelian
(16, 12)	$Q_8 \times C_2$	16	16	nonabelian
(16, 13)	$C_4 \circ D_4$ (Pauli)	16	16	nonabelian

7.2 SymPy verification

We constructed each of the nine groups of Table 4 directly as a regular permutation group on 16 points, using its right regular representation. For every G in the table, `PermutationGroup.centralizer` (applied with S_{16} as the ambient group) returned a group of order 16, and `is_abelian` returned `False`. We additionally computed $|N_G(G_1)|$ by an explicit conjugation test and verified $|C_{S_{16}}(G)| = |N_G(G_1)/G_1|$ in every case, providing an independent check of Proposition 3.1.

7.3 Nonregular exceptions at $d = 16$

A full SageMath enumeration of `TransitiveGroups(16)` identifies exactly 16 nonregular transitive subgroups $G \leq S_{16}$ for which $N_G(G_1)/G_1$ is nonabelian. Adding these to the 9 regular exceptions of Table 4 gives the total $N_{16} - A_{16} = 25$ recorded in Remark 7.2. Every one of the 16 SN cases at

$d = 16$ has $|G_1| > 1$ and $N_G(G_1)/G_1 \cong S_3$ or D_4 , mirroring the S_3 -symmetry observed at $d = 12$ (Section 6) but now with a richer set of underlying G . In contrast to the situation at $d = 12$, where the SN mechanism produces only half as many exceptions as the regular one (3 vs. 3), at $d = 16$ the SN mechanism strictly dominates (16 vs. 9); the gap widens dramatically at $d = 24$ (151 vs. 12, Table 2). A clean group-theoretic classification of the SN exceptions is the content of Question Q1 in Section 8.

8 The Prime-Degree Theorem and Question Q4

We single out one open problem (Question Q4 of the original draft) and provide the structural ingredient that reduces it to a purely Sylow-theoretic statement. Two further open problems (Q1 and Q5) are recorded at the end of the section.

8.1 A primitivity lemma for prime degree

Lemma 8.1. *Let p be a prime and let $G \leq S_p$ be transitive. Then G is primitive.*

Proof. A block of imprimitivity $B \subseteq \Omega$ has size $|B|$ dividing $|\Omega| = p$; since p is prime, $|B| \in \{1, p\}$. The case $|B| = 1$ corresponds to the partition into singletons, and $|B| = p$ corresponds to the partition $\{\Omega\}$. Both are the trivial block systems. Hence G admits no nontrivial block system and is, by Definition 2.5, primitive. $\square$

Theorem 8.2. *Let p be a prime and let $G \leq S_p$ be transitive. Then $C_{S_p}(G)$ is abelian. More precisely, $C_{S_p}(G)$ is either trivial or cyclic of order p .*

Proof. By Lemma 8.1, G is primitive, so Proposition 3.5 gives that $C_{S_p}(G)$ is abelian. The orbits of $C_{S_p}(G)$ on Ω form blocks for G (the centraliser of a transitive group acts semiregularly, Definition 2.4, and its orbits are G -invariant); by primitivity these orbits are either all singletons — in which case $C_{S_p}(G) = \{1\}$ — or a single orbit of length p , in which case $C_{S_p}(G)$ is a transitive semiregular, hence regular, subgroup of S_p of order p , and therefore cyclic of order p (the only group of order a prime). $\square$

Theorem 8.2 explains the experimental observation $A_p = N_p$ for $p \in \{5, 7, 11, 13, 17, 19, 23\}$ (Observation 5.1); the empirical data are tight, in the sense that the centraliser of every transitive group at prime degree is one of the two possibilities (trivial or cyclic of order p) named in the theorem.

Question Q4 of the original draft (“Is Observation 5.1 actually a theorem for all prime d ?”) is therefore answered *in the abelian direction* by Theorem 8.2. The remaining open part of Q4 is the finer classification:

Question Q4 (refined). Determine, for each transitive $G \leq S_p$, which of the two cases $C_{S_p}(G) = \{1\}$ or $C_{S_p}(G) \cong C_p$ occurs, and characterise the latter group-theoretically (e.g. as those G that contain a regular cyclic subgroup of order p).

8.2 Other open questions retained for completeness

(Q1) Can the nonregular exceptions (ids 16, 17, 35 at $d = 12$; the $d = 16$ cases under investigation) be classified into an infinite family? The recurrence of S_3 as $N_G(G_1)/G_1$ suggests a uniform description.

(Q5) Refined in light of the verified data $A_4 = N_4$, $A_9 = N_9$, $A_{25} = N_{25}$ (Observation 5.2): Is it true that $A_{p^2} = N_{p^2}$ for every odd prime p ? The contrast with $d = 16 = 2^4$ ($A_{16}/N_{16} = 1929/1954$) shows that the relevant property is *odd*-prime squared rather than perfect-power; the next test case is $d = 49 = 7^2$ ($N_{49} = ?$ – not yet in the standard libraries within the optimised pipeline’s memory budget).

A Computational artefacts

All code accompanying this draft is in the same directory as the source:

- `sage_verify.sage` — the optimised SageMath driver (Listing 2) producing `summary.csv` and per-degree exception tables.
- `sympy_verify.py` — the independent SymPy re-check for the regular cases at $d = 12, 16$.

Table 5 records the actual wall-times measured on a single laptop core (the same machine that produced Table 2). The total wall-time for the entire sweep $4 \leq d \leq 30$ is under six minutes, dominated by the single row $d = 24$.

Table 5: Wall-time of the optimised pipeline (Listing 2) on a single 2024-era laptop core, per degree.

d range	wall-time (improved pipeline, single laptop core)
$4 \leq d \leq 15$	< 4 s total
$d = 16$	11.1 s
$d = 17, 19, 23, 29$	< 10 ms each (instant via Thm. 8.2)
$d = 18$	6.3 s
$d = 20$	8.8 s
$d = 21, 22, 25, 26$	0.4–1.2 s each
$d = 24$	225.2 s (≈ 3.8 min)
$d = 27$	17.2 s
$d = 28$	17.4 s
$d = 30$	56.3 s
$d = 32$	out of reach (memory: $\sim 2.8 \times 10^6$ classes)

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